Code pièce en T

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This OpenSCAD code generates a hollow T-shaped pipe by defining an outer solid structure and subtracting an inner volume. It starts by setting key parameters such as the outer diameter, wall thickness, and length of the tubes. The exterior module creates the solid T-shape by combining two perpendicular cylinders: one horizontal and one vertical, positioned and rotated to intersect at the center. The interior module defines the hollow part by creating slightly smaller cylinders (accounting for wall thickness) in the same arrangement, with a small length offset to ensure clean subtraction. Finally, the difference operation subtracts the inner volume from the outer body, resulting in a hollow T-shaped piece with uniform wall thickness.

```
// Parameters
outer_diameter = 30;
thickness = 2;
length = 40;


// Outer body (solid T-shape)
module exterior() {
  union() {
    // Horizontal tube
    cylinder(h = length, d = outer_diameter, center = true);


    // Vertical tube
    translate([0, 0, length/2])
      rotate([90, 0, 0])
        cylinder(h = length, d = outer_diameter, center = true);
  }
}
```

```
// Inner volume (to subtract)
module interior() {
  union() {
    // Horizontal hollow
    cylinder(h = length + 2, d = outer_diameter - 2*thickness, center = true);

    // Vertical hollow
    translate([0, 0, length/2])
      rotate([90, 0, 0])
        cylinder(h = length + 2, d = outer_diameter - 2*thickness, center = true);
  }
}

// Final hollow part
difference() {
  exterior();
  interior();
}
```